

In-Depth Review: Deep Threshold-Optimal Policy for MDPs and RMABs

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Outline of Discussion

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- 2 MDP Formulation & State Decoupling
- 3 The Threshold Policy Gradient Theorem
- 4 DeepTOP-MDP Algorithm
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The Inefficiency of Generic RL in Monotone Environments

- Standard Deep RL algorithms (like DDPG or TD3) map states to actions via a generalized neural network $\pi_{\theta}(s)$.
- They treat every state independently. If the temperature is 75° and the AC turns on, standard RL does not inherently "know" that at 76° the AC should also be on.
- **The Monotone Property:**
 - In many control systems (inventory, scheduling, smart grids), optimal actions monotonically depend on a scalar trigger.
 - If action $a = 1$ is optimal at state λ , it remains optimal for all $\lambda' > \lambda$ (or $\lambda' < \lambda$).
- **The Authors' Goal:** Hardcode this inductive bias into the RL architecture to massively reduce the exploration space.

Rigorous MDP Formulation

- The environment is an MDP $\mathcal{E} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathbb{P}, \gamma)$.
- **Action Space:** Binary, $\mathcal{A} := \{0, 1\}$.
- **State Space Decoupling:** $s_t \in \mathcal{S}$ is factored into $s_t = (\lambda_t, v_t)$.
 - ① **Scalar state** $\lambda_t \in \mathbb{R}$: The continuous trigger variable (e.g., current price, current inventory).
 - ② **Vector state** $v_t \in \mathcal{V}$: The discrete contextual variables (e.g., time to deadline, seasonality).
- **Threshold Function:** The policy learns a mapping $\mu : \mathcal{V} \rightarrow \mathbb{R}$.
- **Action Execution:** $a_t = \mathbb{I}(\mu(v_t) > \lambda_t)$.

The Continuous Objective Function

- Let $\mu^\phi(v)$ be our threshold function parameterized by weights ϕ .
- Let $Q_{\mu^\phi}(\lambda, v, a)$ be the expected discounted reward from taking action a in state (λ, v) and following μ^ϕ thereafter.
- We want to find ϕ that maximizes the expected reward across all possible states.
- The global objective function integrates over the entire bounded continuous space $[-M, +M]$ of the scalar state λ :

$$K(\mu^\phi) := \int_{\lambda=-M}^{+M} \sum_{v \in \mathcal{V}} Q_{\mu^\phi}(\lambda, v, \mathbb{I}(\mu^\phi(v) > \lambda)) d\lambda \quad (1)$$

The Mathematical Bottleneck: Bypassing the Integral

- We need to compute $\nabla_{\phi} K(\mu^{\phi})$.
- The indicator function $\mathbb{I}(\mu^{\phi}(v) > \lambda)$ creates discontinuous jumps in the action space.
- **The Authors' Derivation Strategy:**
 - 1 Use the Fubini-Tonelli theorem to swap summation and integration.
 - 2 Break the integral into discrete intervals bounded by the thresholds of different vector states.
 - 3 Apply the **Leibniz Integral Rule** to differentiate under the integral sign.

The Threshold Policy Gradient Theorem

Theorem 1 (Gradient Expression)

If all vector states $v \in \mathcal{V}$ have distinct values of $\mu^\phi(v)$, the gradient of the objective function simplifies completely to:

$$\nabla_\phi K(\mu^\phi) = 2M|\mathcal{V}| \sum_{v \in \mathcal{V}} \bar{\rho}(\mu^\phi(v), v) \underbrace{\left[Q_{\mu^\phi}(\mu^\phi(v), v, 1) - Q_{\mu^\phi}(\mu^\phi(v), v, 0) \right]}_{\text{Advantage at the Boundary}} \nabla_\phi \mu^\phi(v)$$

- $\bar{\rho}(\cdot)$ is the discounted state visitation distribution.
- The integral disappears entirely!

DeepTOP-MDP: Actor-Critic Architecture

- To implement Theorem 1, the authors build a specialized Actor-Critic architecture.
- **Critic Network** $Q^\theta(\lambda, v, a)$:
 - Takes the full state (λ, v) and action a as input.
 - Trained via standard Temporal Difference (TD) learning (like in DDPG/TD3).
- **Actor Network** $\mu^\phi(v)$:
 - Takes ONLY the vector state v as input.
 - Outputs a single continuous value (the threshold).
 - Trained using the specialized gradient from Theorem 1.

DeepTOP-MDP: Minibatch Update Rules

- From a replay buffer $\mathcal{M}$, sample a minibatch of transitions $\{s_{t_k}, a_{t_k}, r_{t_k}, s_{t_k+1}\}$.
- **Critic Loss Gradient:**

$$\hat{\nabla}_{\theta} \mathcal{L}(\theta) = \frac{2}{B} \sum_{k=1}^B \left(Q_k^{\theta} - r_{t_k} - \gamma \max_{a'} Q_{k+1}^{\theta'} \right) \nabla_{\theta} Q_k^{\theta}$$

- **Actor Gradient Estimation (Empirical Theorem 1):**

$$\hat{\nabla}_{\phi} K(\mu^{\phi}) = \frac{1}{B} \sum_{k=1}^B \left(Q^{\theta}(\mu^{\phi}(v_{t_k}), v_{t_k}, 1) - Q^{\theta}(\mu^{\phi}(v_{t_k}), v_{t_k}, 0) \right) \nabla_{\phi} \mu^{\phi}(v_{t_k})$$

The Curse of Restless Multi-Armed Bandits (RMABs)

- An RMAB system has N independent Markovian arms.
- In every timestep, the agent must activate exactly V arms out of N .
- **The Curse of Dimensionality:** The state space grows exponentially as $\mathcal{S}_1 \times \mathcal{S}_2 \times \cdots \times \mathcal{S}_N$.
- **The Standard Heuristic:** The Whittle Index Policy.
 - Assign an independent "index" $W_i(s_{i,t})$ to each arm based on its current state.
 - Sort the arms by their indices.
 - Activate the top V arms.

Reframing the Whittle Index as a Threshold Policy

- The Whittle index can be computed by looking at a **single arm in isolation** under an alternative problem formulation.
- **The Alternative Problem:**
 - The agent decides whether to activate arm i .
 - If activated, the agent pays an artificial *activation cost* λ .
 - Net reward: $r_{i,t} - \lambda a_{i,t}$.
- A threshold policy activates the arm iff $\mu_i(s_{i,t}) > \lambda$.
- **The Insight:** The value $\mu_i(s_{i,t})$ represents the maximum cost the agent is willing to pay. *This threshold is exactly the Whittle index $W_i(s_i)$.*

The DeepTOP-RMAB Algorithm

- We maintain separate actor (ϕ_i) and critic (θ_i) networks for **each** arm i .
- Because λ is an artificial construct, we simulate it during the update step!
- **Training loop for arm i :**
 - ① Sample physical transitions $\{s_{i,t_k}, a_{i,t_k}, r_{i,t_k}, s_{i,t_k+1}\}$ from memory.
 - ② **Randomly sample B values of artificial cost λ from $[-M, +M]$.**
 - ③ Update the critic $Q_i^{\theta_i}$ to learn the value of the arm given state and cost λ .
 - ④ Update the actor $\mu_i^{\phi_i}$ using the exact same gradient expression from Theorem 1.

Experimental Setup: MDP Environments

The authors tested DeepTOP against generic Deep RL (DDPG, TD3) and SALMUT in 3 highly specific environments:

① Electric Vehicle (EV) Charging:

- *Scalar*: Stochastic electricity price (Ornstein-Uhlenbeck process).
- *Vector*: Remaining charge needed, time to deadline.
- *Goal*: Fulfill charge requirement before deadline at lowest cost.

② Inventory Management:

- *Scalar*: Current inventory level.
- *Vector*: Seasonal demand index (Poisson rates dependent on season).

③ Make-to-Stock Production:

- Servicing orders from W demand classes with varying rewards and a finite buffer.

Results: MDP Environments

- **Performance:** DeepTOP significantly outperforms DDPG, TD3, and SALMUT in average reward and convergence speed.
- **Why standard RL failed:** DDPG and TD3 suffer high variance because they attempt to learn the optimal policy for each continuous scalar state independently.
- **Optimality Check:** For EV charging, the mathematical optimal policy ("Deadline Index") is known. DeepTOP converged to the exact theoretical optimal policy in just 2000 timesteps.

Experimental Setup: RMAB Environments

Tested against Neural LPQL, Neural WIBQL, and NeurWIN across varying dimensions (e.g., $N = 30$, $V = 6$).

① One-Dimensional Bandits:

- Each arm is a 100-state Markov process.
- Reward is based on proximity to state 99.
- Activation pushes the state $+1$; passive drops the state -1 .

② Recovering Bandits:

- Models consumer ad-click behavior.
- State is "time since last click."
- Reward follows a recovering exponential function (interest drops upon clicking, then slowly recovers).

Results: RMAB Environments

- **DeepTOP Dominance:** DeepTOP achieved the highest optimal average rewards across all combinations of N and V .
- **Failure of Baselines:**
 - *Neural LPQL*: Relies on computing a common Lagrangian across all arms. An inaccuracy in just one arm's Q-network poisons the global index calculation.
 - *NeurWIN*: Based on REINFORCE (Monte Carlo). It suffers from high variance and can only apply gradient updates at the end of full episodes.
- **DeepTOP Advantage:** By training independently via Temporal Difference learning, DeepTOP isolates errors and learns from step-by-step transitions.

Summary of Contributions

- 1 **Theoretical:** Developed the Threshold Policy Gradient Theorem, proving that continuous integrals over threshold boundaries simplify into easily computable edge-advantages.
- 2 **Algorithm (MDP):** Proposed DeepTOP-MDP, which hardcodes the monotone inductive bias into an actor-critic framework, vastly outpacing generic Deep RL.
- 3 **Algorithm (RMAB):** Proved the Whittle Index maps perfectly to this framework via an artificial activation cost, leading to DeepTOP-RMAB.
- 4 **Empirical:** Demonstrated state-of-the-art performance in complex scheduling, inventory, and bandit environments.

- **Multiple Actions:** The current framework is strictly binary (action 0 or 1).
- How do we extend this to continuous or multi-level actions?
- *Example:* The Federal Reserve deciding not just *if* to raise interest rates, but *by how much* (e.g., 25 bps vs 50 bps).
- **Indexability Constraints:** While DeepTOP-RMAB finds local optima, theoretical guarantees of convergence to the true Whittle Index only hold if the underlying arms are strictly indexable.

Questions or Discussion?

Next Steps: Stress-Testing DeepTOP on Complex RMABs

Motivation: The RMAB environments evaluated in the paper are relatively simple. How robust is DeepTOP against structural violations?

- **Non-indexable MDPs:** Introduce "reward cliffs" where the optimal policy is non-monotone. Will DeepTOP's actor converge to a theoretically unsound proxy?
- **Multi-dimensional States:** Expand beyond 1D states to assess if the sample complexity grows prohibitively.
- **Correlated Arms:** Violate the independent Lagrangian decoupling assumption (e.g., shared queuing buffers or epidemic spread).
- **Non-stationary Dynamics:** Implement drifting transition probabilities (p, q) to test if the critic can adapt without explicit detection mechanisms.
- **Sparse & Delayed Rewards:** Withhold shaped rewards (e.g., only rewarding at state 99) to test critic variance over long effective horizons.
- **Tight Budgets at Scale:** Push to extreme ratios (e.g., $N = 30, V = 1$) where the uniform price sampling calibration fails and a single mis-ranking is catastrophic.